Program DipoleL.m

Syntax:

```
[v,pc,ps] = DipoleL(theta,z,n1,n,n2,d1,d,d2)
```

Description:

DipoleL calculates the angular distribution of emission into the lower half space of an oscillating dipole at a distance z from the bottom interface of a layer with refractive index n and thickness d that is sandwiched between two stacks of layers with refractive indices $n_{1,1}$ through $n_{1,k1}$ and thickness values $d_{1,1}$ through $d_{1,k1-1}$ (bottom stack, from bottom to top) and refractive indices $n_{2,1}$ through $n_{2,k2}$ and thickness values $d_{2,1}$ through $d_{2,k2-1}$ (top stack, from bottom to top).

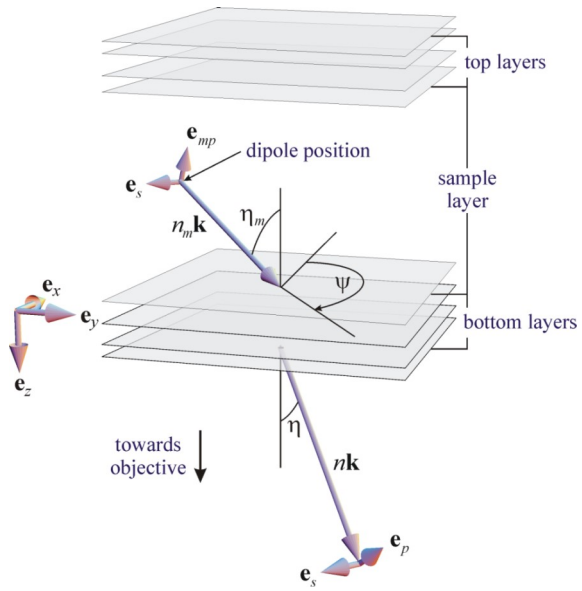


Fig.1: Geometry of he sample space: the sample solution is stacked between stacks of layers with homogeneous optical properties.

Input variables: (all length values in units of emission wavelength over 2π)

theta	–	polar emission angle
z	–	molecule's distance from the bottom of its layer
n1	–	vector of refractive indices of the stack below the molecule's layer
n	–	refractive index of the molecule's layer
n2	–	vector of refractive indices of the stack above the molecule's layer
d1	–	vector of layer thickness values of the stack below the molecule's layer (length(d1) = length(n1)-1)
d	–	thickness of molecule's layer
d2	–	vector of layer thickness values of the stack above the molecule's layer (length(d2) = length(n2)-1)

Output variables (all emission powers are in such units so that a free dipole within a homogeneous medium with refractive index *n* has an emission power of 4/3·*n*):

v – Angular distribution of electric field amplitude of a vertically oriented dipole (perpendicular to interfaces) into lower half-space;

pc,ps – *p*- and *s*-wave components of the angular distribution of electric field amplitude for a horizontally oriented dipole (parallel to interfaces) into lower half-space. The *p*-wave component (**pc**) varies as cos(ϕ), and the *s*-wave component (**ps**) as sin(ϕ), where ϕ is the azimuthal emission angle, ϕ = 0 being the orientation of the in-plane projection of the dipole's orientation.

Example 1:

Consider the emission of a dipole in air close to a glass/air interface. The refractive index of air is 1.0, that of glass 1.51. Emission wavelength is 570 nm. Let us consider two different dipole positions, z = 0 nm, and 570 nm. First, let us calculate the emission into the glass (lower half space). The input parameters are

```
lam = 570;
ng = 1.51; n = 1.0;
d1 = []; d = 570/lam*2*pi; d2 = [];
z = (0:1)*2*pi;
theta = (0:1e3)'/1e3*pi/2;
```

and the calculation is done by

```
for j=1:2
    [v1(:,j), pc1(:,j), ps1(:,j)] = ...
          DipoleL(theta,z(j),n1,n,n,d1,d,d2);
end
```

Next, the emission into the air half space

```
for j=1:2
    [v2(:,j), pc2(:,j), ps2(:,j)] = ...
          DipoleL(theta,d-z(j),n,n,n1,d1,d,d2);
end
```

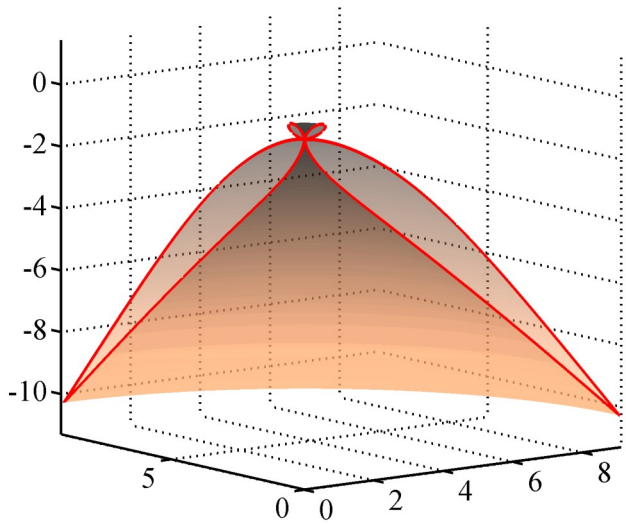
Let us plot the angular distribution of emission for a vertical dipole directly on the glass surface. First, we define the azimuthal variable ϕ:

```
phi = (0.5:90)/360*2*pi;
```

Then, we make a 3d surface plot for the vertical dipole directly on the surface:

```
j = 1;
surf((abs(v1(:,j)).^2.*sin(theta))*cos(phi),(abs(v1(:,j)).^2.*sin(t
heta))*sin(phi),-(abs(v1(:,j)).^2.*cos(theta))*ones(size(phi)),
abs(v1(:,j)).^2*ones(size(phi)), 'faceAlpha',0.5);
hold on
```

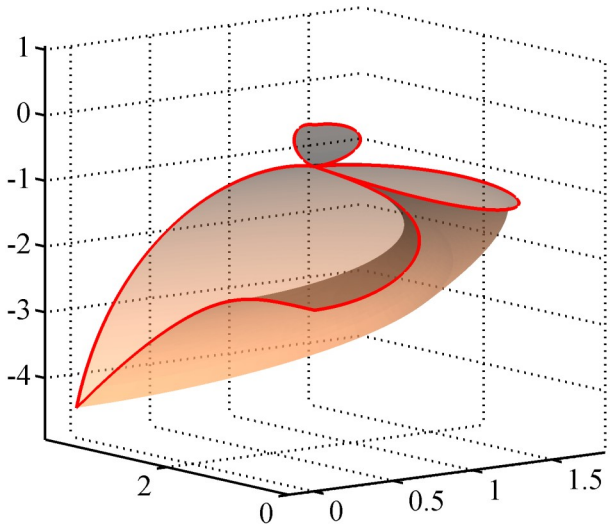
```
surf((abs(v2(:,j)).^2.*sin(theta))*cos(phi),
(abs(v2(:,j)).^2.*sin(theta))*sin(phi),(abs(v2(:,j)).^2.*cos(theta)
)*ones(size(phi)), abs(v2(:,j)).^2*ones(size(phi)),
'faceAlpha',0.5);
plot3((abs(v1(:,j)).^2.*sin(theta))*cos(phi(1)),(abs(v1(:,j)).^2.*s
in(theta))*sin(phi(1)),-abs(v1(:,j)).^2.*cos(theta), 'r');
plot3((abs(v1(:,j)).^2.*sin(theta))*cos(phi(end)),(abs(v1(:,j)).^2.
*sin(theta))*sin(phi(end)),-abs(v1(:,j)).^2.*cos(theta), 'r');
plot3((abs(v2(:,j)).^2.*sin(theta))*cos(phi(1)),(abs(v2(:,j)).^2.*s
in(theta))*sin(phi(1)), abs(v2(:,j)).^2.*cos(theta), 'r');
plot3((abs(v2(:,j)).^2.*sin(theta))*cos(phi(end)),(abs(v2(:,j)).^2.
*sin(theta))*sin(phi(end)), abs(v2(:,j)).^2.*cos(theta), 'r');
hold off
colormap copper
shading flat
axis('vis3d')
view([-37.5 10])
```



And the same for a horizontal dipole:

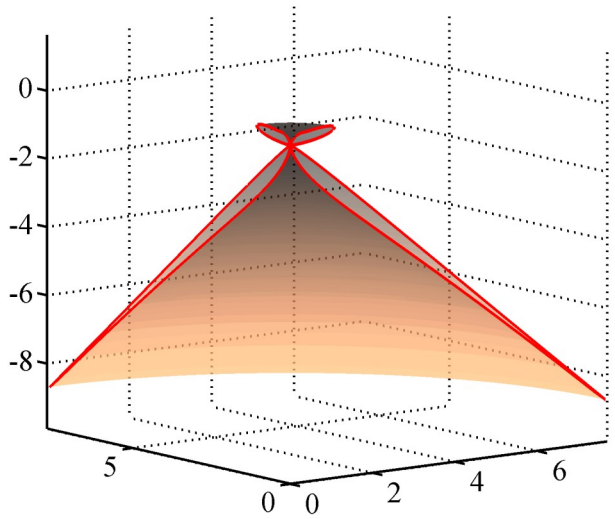
```
j = 1;
f1 = abs(pc1(:,j)*cos(phi)).^2+abs(ps1(:,j)*sin(phi)).^2;
surf(f1.*(sin(theta)*cos(phi)),f1.*(sin(theta)*sin(phi)),-
f1.*(cos(theta)*ones(size(phi))), f1, 'faceAlpha',0.5);
hold on
f2 = abs(pc2(:,j)*cos(phi)).^2+abs(ps2(:,j)*sin(phi)).^2;
surf(f2.*(sin(theta)*cos(phi)),f2.*(sin(theta)*sin(phi)),
f2.*(cos(theta)*ones(size(phi))), f2, 'faceAlpha',0.5);
plot3(f1(:,1).*(sin(theta)*cos(phi(1))),
f1(:,1).*(sin(theta)*sin(phi(1))), -f1(:,1).*cos(theta), 'r');
plot3(f1(:,end).*(sin(theta)*cos(phi(end))),
f1(:,end).*(sin(theta)*sin(phi(end))), -f1(:,end).*cos(theta),
'r');
plot3(f2(:,1).*(sin(theta)*cos(phi(1))),f2(:,1).*(sin(theta)*sin(ph
i(1))), f2(:,1).*cos(theta), 'r');
```

```
plot3(f2(:,end).*(sin(theta)*cos(phi(end))),f2(:,end).*(sin(theta)*
sin(phi(end))), f2(:,end).*cos(theta), 'r');
hold off
colormap copper
shading flat
axis('vis3d')
view([-37.5 10])
```



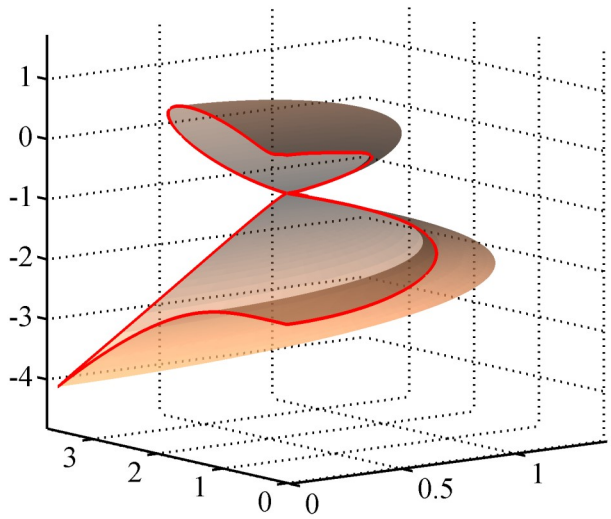
Now, we move the dipole one wavelength away from the glass surface:

```
j = 2;
surf((abs(v1(:,j)).^2.*sin(theta))*cos(phi),(abs(v1(:,j)).^2.*sin(t
heta))*sin(phi),-(abs(v1(:,j)).^2.*cos(theta))*ones(size(phi)),
abs(v1(:,j)).^2*ones(size(phi)), 'faceAlpha',0.5);
hold on
surf((abs(v2(:,j)).^2.*sin(theta))*cos(phi),
(abs(v2(:,j)).^2.*sin(theta))*sin(phi),(abs(v2(:,j)).^2.*cos(theta)
)*ones(size(phi)), abs(v2(:,j)).^2*ones(size(phi)),
'faceAlpha',0.5);
plot3((abs(v1(:,j)).^2.*sin(theta))*cos(phi(1)),(abs(v1(:,j)).^2.*s
in(theta))*sin(phi(1)),-abs(v1(:,j)).^2.*cos(theta), 'r');
plot3((abs(v1(:,j)).^2.*sin(theta))*cos(phi(end)),(abs(v1(:,j)).^2.
*sin(theta))*sin(phi(end)),-abs(v1(:,j)).^2.*cos(theta), 'r');
plot3((abs(v2(:,j)).^2.*sin(theta))*cos(phi(1)),(abs(v2(:,j)).^2.*s
in(theta))*sin(phi(1)), abs(v2(:,j)).^2.*cos(theta), 'r');
plot3((abs(v2(:,j)).^2.*sin(theta))*cos(phi(end)),(abs(v2(:,j)).^2.
*sin(theta))*sin(phi(end)), abs(v2(:,j)).^2.*cos(theta), 'r');
hold off
colormap copper
shading flat
axis('vis3d')
view([-37.5 10])
```



and

```
j = 2;
f1 = abs(pc1(:,j)*cos(phi)).^2+abs(ps1(:,j)*sin(phi)).^2;
surf(f1.*(sin(theta)*cos(phi)),f1.*(sin(theta)*sin(phi)),-
f1.*(cos(theta)*ones(size(phi))), f1, 'faceAlpha',0.5);
hold on
f2 = abs(pc2(:,j)*cos(phi)).^2+abs(ps2(:,j)*sin(phi)).^2;
surf(f2.*(sin(theta)*cos(phi)),f2.*(sin(theta)*sin(phi)),
f2.*(cos(theta)*ones(size(phi))), f2, 'faceAlpha',0.5);
plot3(f1(:,1).*(sin(theta)*cos(phi(1))),
f1(:,1).*(sin(theta)*sin(phi(1))), -f1(:,1).*cos(theta), 'r');
plot3(f1(:,end).*(sin(theta)*cos(phi(end))),
f1(:,end).*(sin(theta)*sin(phi(end))), -f1(:,end).*cos(theta),
'r');
plot3(f2(:,1).*(sin(theta)*cos(phi(1))),f2(:,1).*(sin(theta)*sin(ph
i(1))), f2(:,1).*cos(theta), 'r');
plot3(f2(:,end).*(sin(theta)*cos(phi(end))),f2(:,end).*(sin(theta)*
sin(phi(end))), f2(:,end).*cos(theta), 'r');
hold off
colormap copper
shading flat
axis('vis3d')
view([-37.5 10])
```



Example 2:

Let us calculate the angular distribution of emission into air of a dipole five wavelengths away from a gold surface at 570 nm. The complex-valued refractive index of gold at 570 nm is 0.3257 + 2.5792·*i*.
Thus, one has

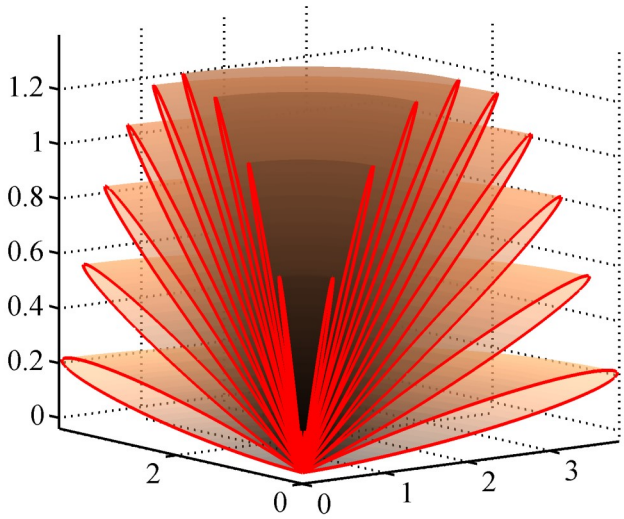
```
lam = 570;
n1 = 0.3257 + 2.5792i; n = 1.0; n2 = 1.0;
d1 = []; d = 10*pi; d2 = [];
theta = (0:1e3)'/1e3*pi/2;
```

and the calculation is done by

```
[v, pc, ps] = DipoleL(theta,0,n,n,n1,d1,d,d2);
```

The result can be plotted similarly to Example 1. First, the vertical dipole:

```
surf((abs(v).^2.*sin(theta))*cos(phi),(abs(v).^2.*sin(theta))*sin(p
hi), (abs(v).^2.*cos(theta))*ones(size(phi)),
abs(v).^2*ones(size(phi)), 'faceAlpha',0.5);
hold on
plot3((abs(v).^2.*sin(theta))*cos(phi(1)),(abs(v).^2.*sin(theta))*s
in(phi(1)),abs(v).^2.*cos(theta), 'r');
plot3((abs(v).^2.*sin(theta))*cos(phi(end)),(abs(v).^2.*sin(theta))
*sin(phi(end)),abs(v).^2.*cos(theta), 'r');
hold off
colormap copper
shading flat
axis('vis3d')
view([-37.5 10])
```


1.2
1
0.8
0.6
0.4
0.2
0
2
0
0
1
2
3